

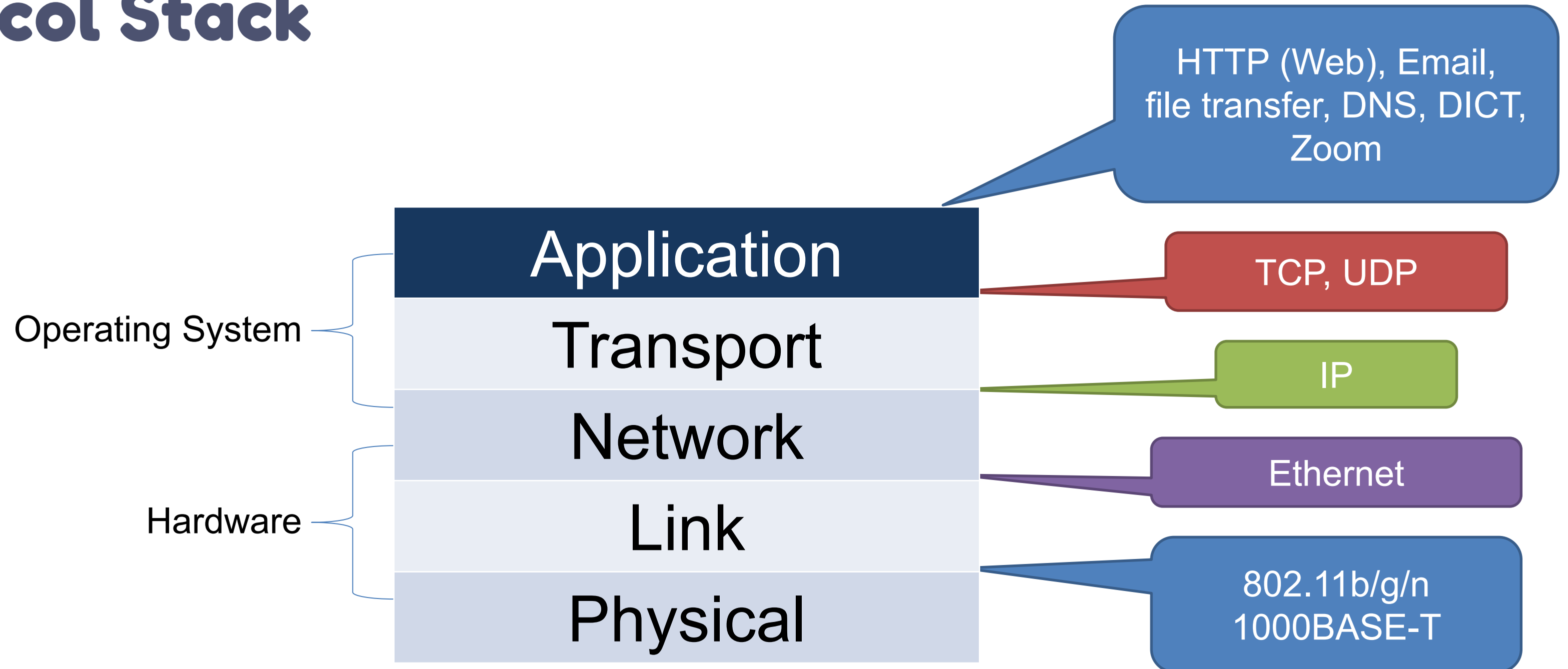
Application Layer Protocols

Module 3.1

READING

- Reading: 2.1, 2.7

Protocol Stack



Learning Goals – General

- Explain design considerations for application protocols
- Explain the advantages and disadvantages of open (defined by a standard) vs. closed (proprietary) protocols
- Explain the difference between a peer-to-peer and a client-server application protocol
- Explain the quality of service requirements for different applications
- Explain what a socket and a transport layer address is
- Effectively use Java APIs or C system calls to create/destroy sockets and send/receive data

Design Considerations For Application-layer Protocols

- Each application using the network will define its own protocol
- Open vs Proprietary
- Architecture: client-server, peer-to-peer (P2P)
 - Who is the client, who is the server
 - How does the client identify which server to contact
 - Rules for when client and/or server send/receive messages
- Choice of transport protocol
 - Desired quality of service
- Types and formats of messages (request, response, etc.)
 - Message syntax and semantics
 - Message encoding format (text, binary, etc.)

Open Vs Proprietary Protocols

- Open protocols: publicly known
 - Examples: DICT, HTTP, SMTP, SSH
 - Usually defined in RFC (Request for Comments) documents
 - Many different implementations
- Proprietary protocols
 - Examples: Skype, iCloud, Zoom
 - Only one implementation

Client-Server Architecture

- Well-defined roles for client and server
- Server is always on, with permanent address or host name
- Client establishes connection
- Connection is always between one client and one server (although the server will serve multiple clients at once)

Peer-to-Peer Architecture

- Connections typically between peers with the same hierarchical role
 - Some hierarchy may be used, but connection is not restricted to it
- Peers request service from other peers, provide service in return
- Self scalability: new peers bring new demand and new capacity
- Complex peer address management

Clicker Question

Which of the following is NOT a characteristic of a peer-to-peer application?

- A. Hosts are intermittently connected
- B. Application is self-scalable
- C. All hosts are known *a priori*
- D. All of the above

Clicker Answer

Which of the following is NOT a characteristic of a peer-to-peer application?

- A. Hosts are intermittently connected
- B. Application is self-scalable
- C. All hosts are known *a priori***
- D. All of the above

What Quality Of Service Does An Application Need?

Data loss

- some apps can tolerate some loss (e.g., audio)
- other apps require 100% reliable data transfer (e.g., file transfer, web, email)

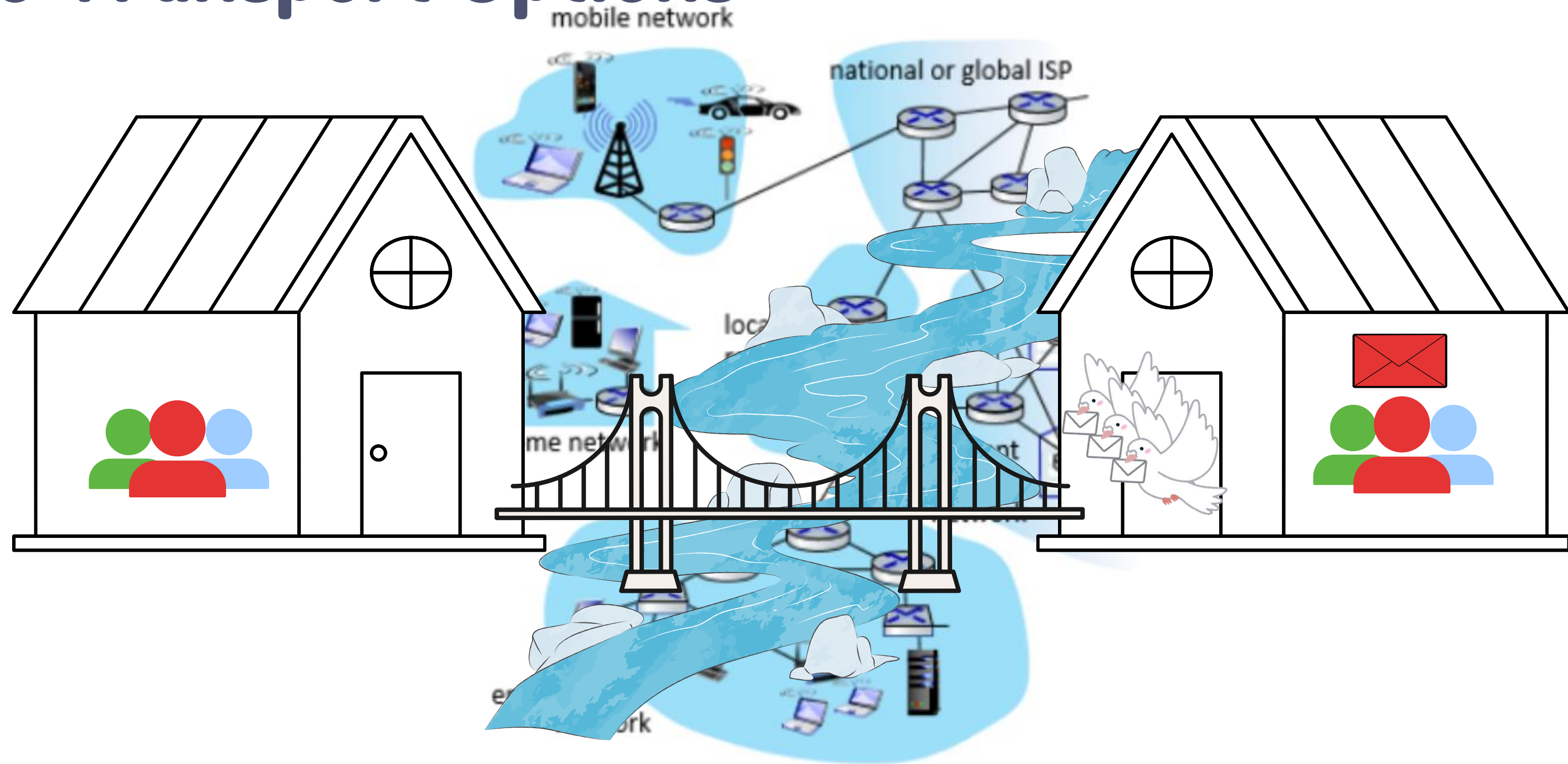
Time sensitivity

- some apps require low delay to be “effective” (e.g., interactive ones)

Bandwidth

- some apps (e.g., multimedia) require minimum amount of bandwidth to be “effective”
- other apps (“elastic apps”) make use of whatever bandwidth they get

Two Transport Options



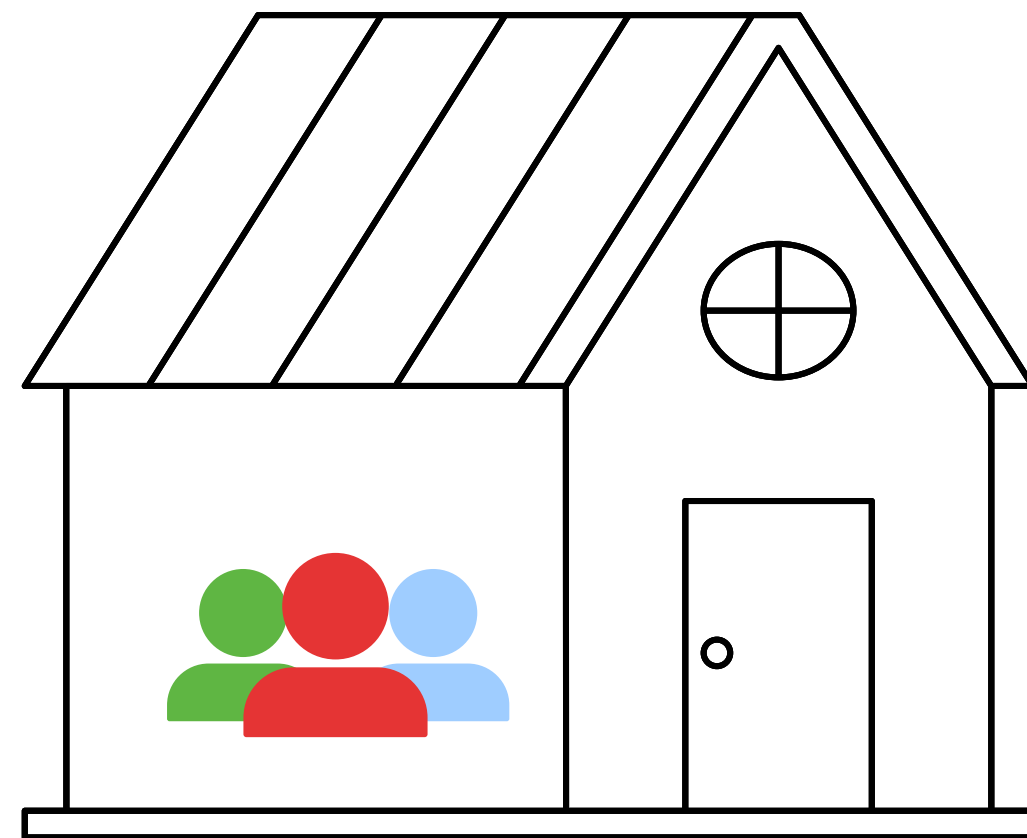
Two Transport Options



Simple
Fast

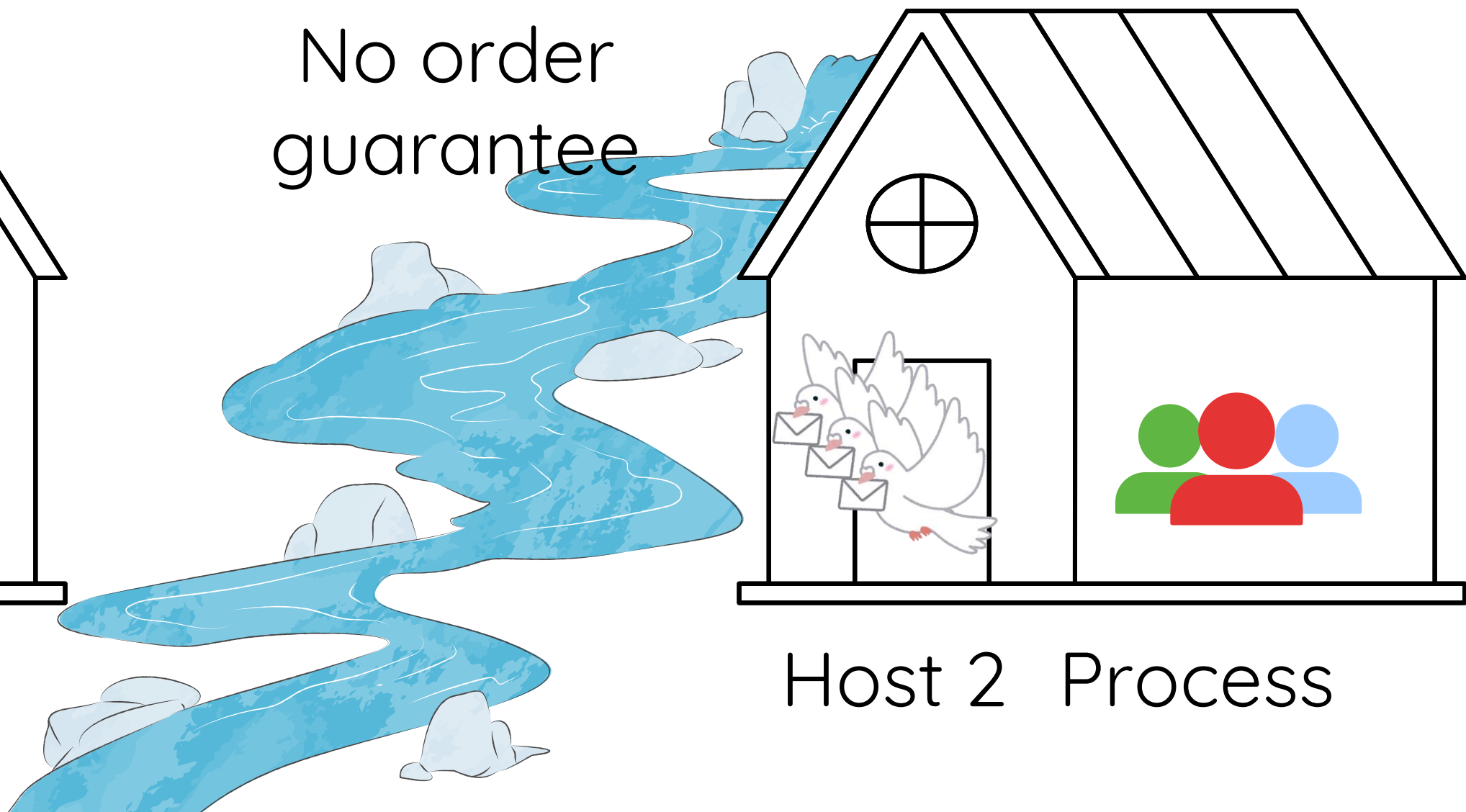


Unreliable
No order
guarantee



Process

Host 1



Host 2 Process

Two Transport Options

✓ Reliable
In-order delivery
Congestion control

✗ Slower



Two Transport Options

Reliable stream	Unreliable packet
Connection	No connection
Reliable ordered delivery	Best effort
Flow/Congestion control	Nope
Possible delays	No (transport level) delay

TCP

UDP

Application Examples

- File transfer, web, email → TCP
 - Loss averse, not time sensitive, elastic bandwidth
- Text messaging → TCP or UDP
 - Loss averse, elastic bandwidth, somewhat time sensitive
- On demand multimedia streaming → UDP or TCP
 - Some loss tolerance, somewhat time sensitive
- Real time multimedia, VoIP, interactive games → UDP
 - Some loss tolerance, time sensitive, bandwidth requirements
- Domain Name Service → UDP
 - Loss tolerant, not time sensitive, elastic bandwidth

The DICT Protocol

- Defined in RFC 2229
 - Google: dict protocol rfc
- Simple text-based, request-response protocol
- Commands: help, define, match, show db, show strat, quit
- Example: netcat dict.org 2628
 - Wait, what is this 2628??

A Digression On Network Transport

- We will talk about the transport layer protocols later (module 4), but ...
- You need to know how applications see the layer below them in the network protocol stack
- Two things you need to be aware of:
 - A transport layer address – how network applications are identified
 - A socket – a network end point

Transport Layer Addresses

- A pair of a 32-bit IP host address and a 16-bit port number
 - or a 128-bit IPv6 host address and a 16-bit port number
- Usually the IP address is derived from a DNS name
 - `www.cs.ubc.ca`, `google.ca`, `amazon.ca`, ...

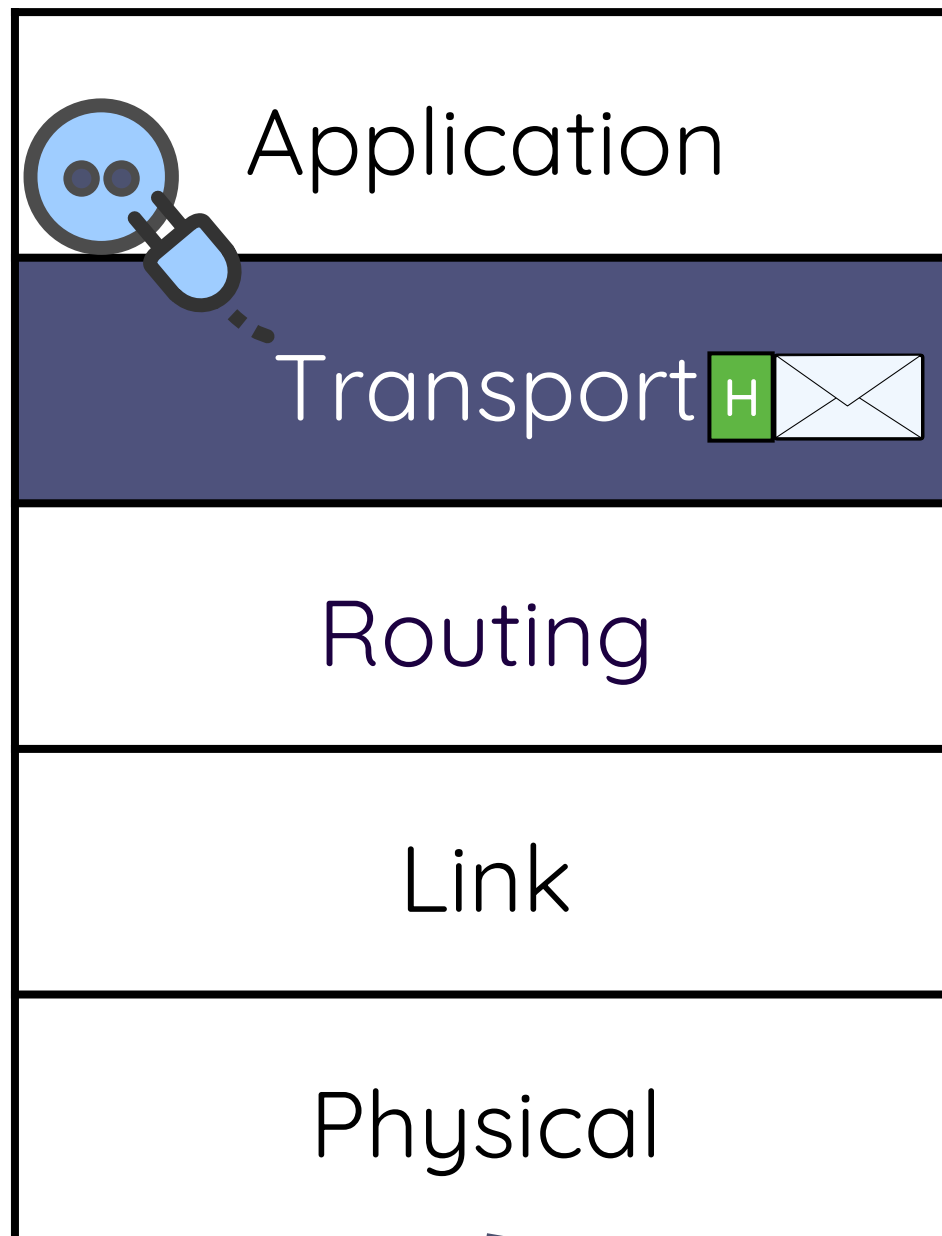
netcat dict.org 2628



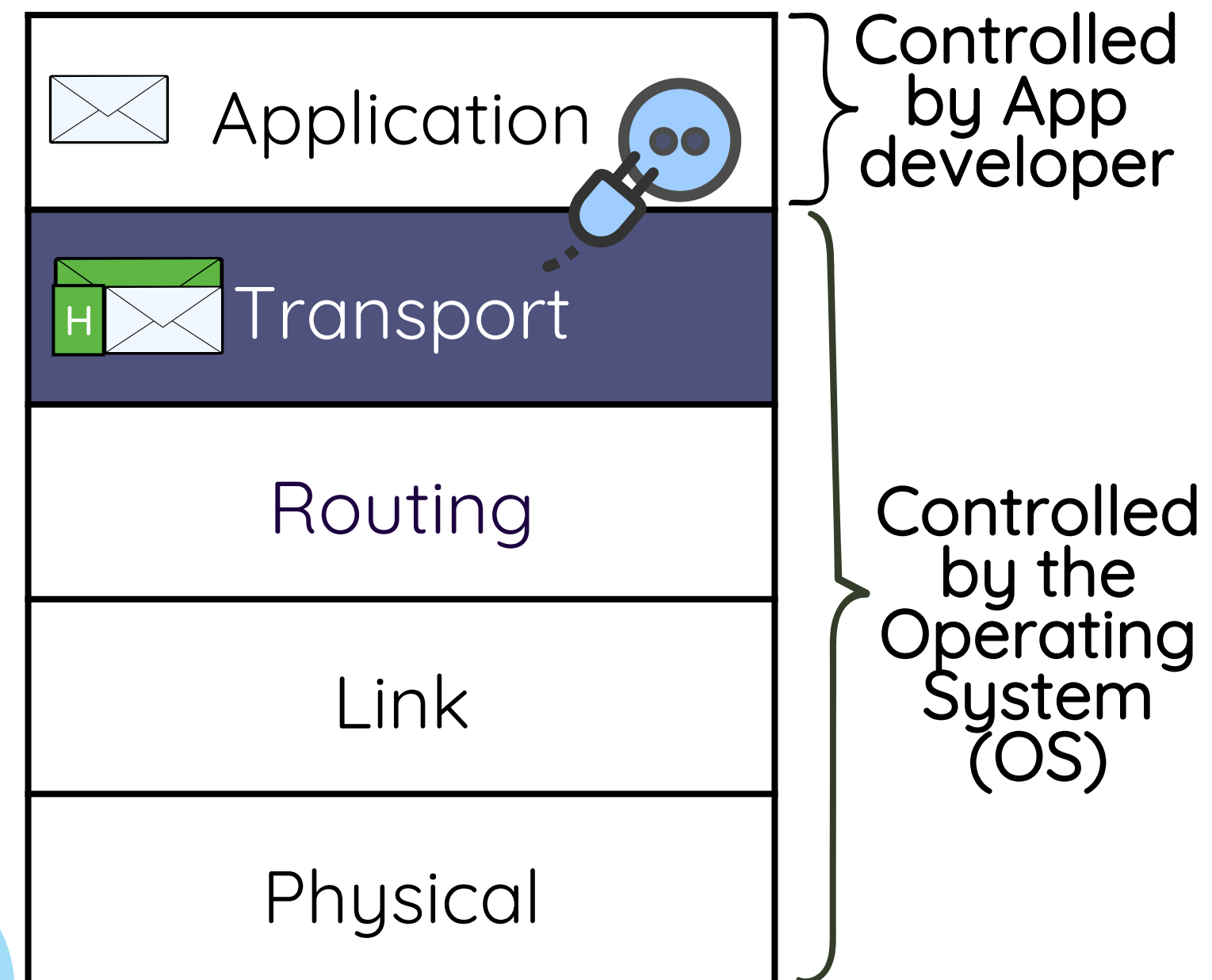
DNS
Name Port
Number

dict.org -> 199.48.130.6

Receiver



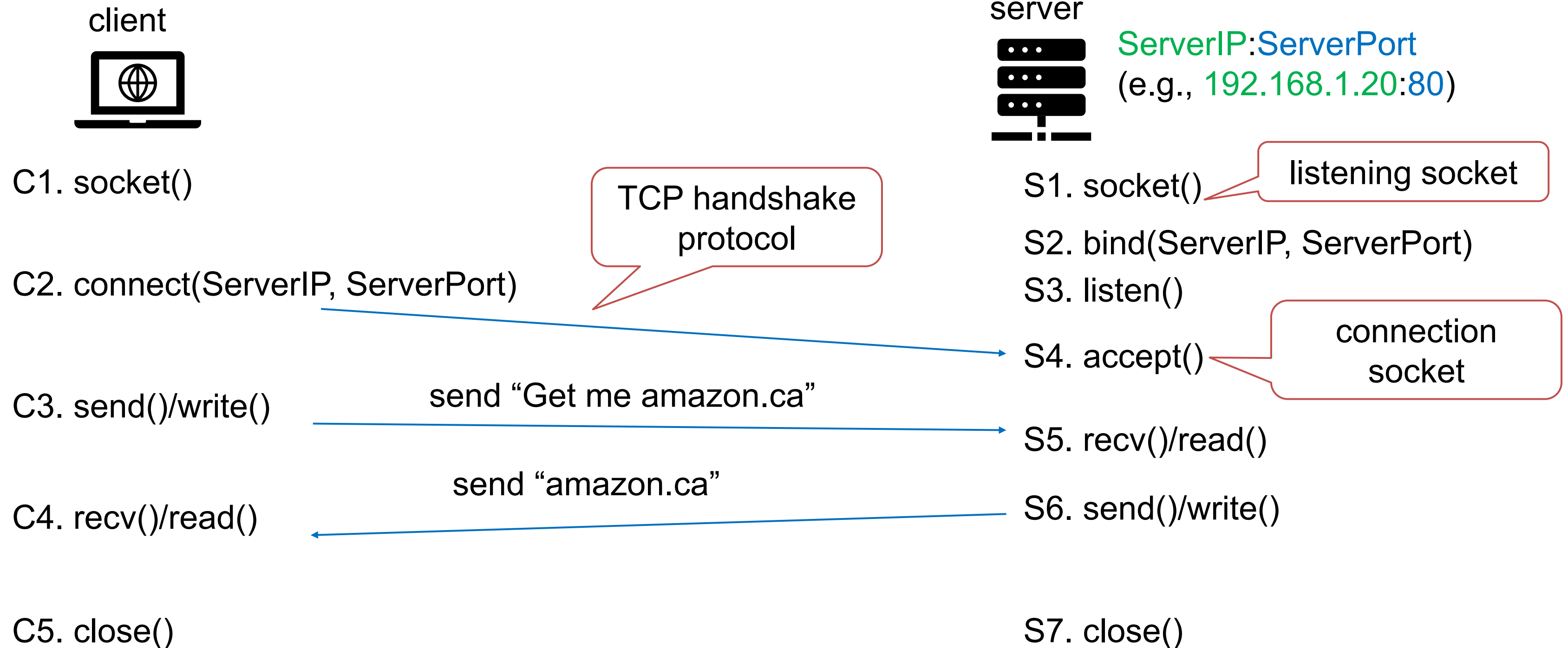
Sender



Sockets – Network Endpoints

- Created via `socket()` system call
 - Parameters to the `socket()` call identify the transport protocol and optionally the other participating process
- Destroyed via `close()`
 - like all other file descriptors
- Data sent and received using `send()` and `recv()`
 - or `read()` and `write()` if the socket is “connected”

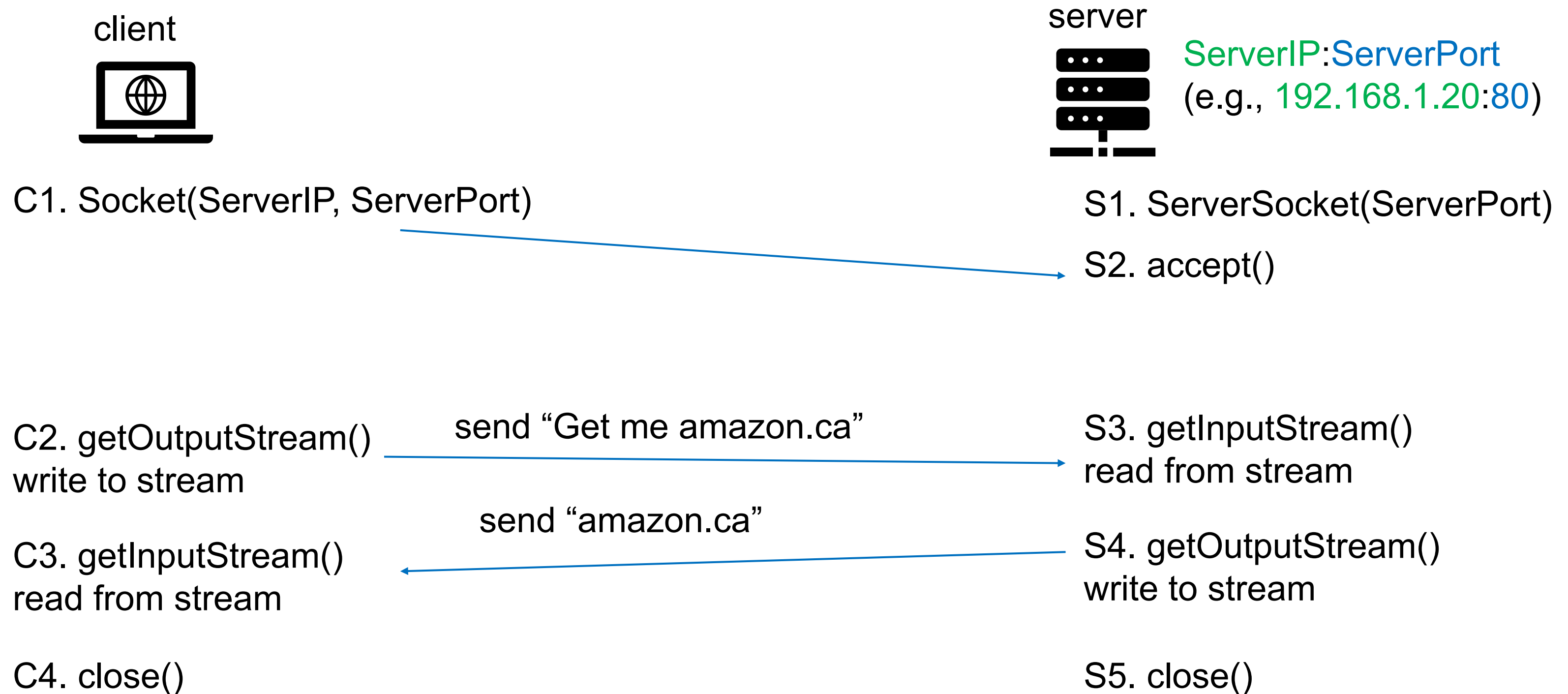
Sockets In C (Tcp Client-Server)



Sockets in Java

- class Socket for the client side
- new Socket(host, port)
 - host is a String
 - port is an int
 - Creates a socket, connects it to the indicated server, and returns the socket
 - Throws various exceptions when things go wrong
- class ServerSocket() for the server side

Sockets In Java (Tcp Client-Server)



The Simplest Java Socket Client

```
private void body() {  
    try (Socket s = new Socket("dict.org", 2628)) {  
        BufferedReader input = new BufferedReader(new InputStreamReader(s.getInputStream()));  
        String greeting = input.readLine();  
        System.out.println(greeting);  
    } catch (IOException e) {  
        e.printStackTrace();  
    }  
}
```

In-class Activity

- Form yourselves into groups (1 – 9 students per group)
- You should be able to chat conveniently in your group
- Go to PrairieLearn
- Click on Assessments
- Start the ICA31 assessment (Application Architecture and Transport Protocols)
- Talk in your group about the answers
 - Hearing other students ideas
 - Explaining your ideas to others

Next Topic: Application Layer – The Web